

GANGASAGAR HL

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EDUCATION

M.Tech in Computer Science, International Institute of Information Technology Bangalore July 2024 – Present
GPA: 3.58/4

Bachelor of Engineering in Computer Science, Sapthagiri College Engineering August 2017 – July 2021
GPA: 3.59/4

EXPERIENCE

IoT Intern, SmartBridge, Remote [Link](#) Aug 2020

- Built end-to-end smart surveillance system using IBM Watson IoT for real-time monitoring and control.
- Integrated image capture with Cloud Object Storage, Cloudant DB, and Watson Visual Recognition for threat detection.
- Orchestrated flows via Node-RED and triggered Fast2SMS mobile alerts for unknown face detection.

PROJECTS

Banking Management System — C [Link](#) November 2024

- Developed a system simulating core banking functionalities including customer management, transaction handling, loan processing, and administrative oversight.
- Implemented using system calls, websockets, and file locking programming concepts.

Security System — Python, Flask, PyTorch, LangChain, TFLite, FAISS, Raspberry Pi 3B+ [Link](#) Apr 2025

- Deployed MobileNetV3-small (TFLite) on Raspberry Pi for real-time detection and alert-based media streaming.
- Integrated BLIP, T5, LLaMA 3.2 for alert summarization with Twilio/WhatsApp/Gmail notifications.
- Built FAISS-based RAG pipeline to index/summarize visual logs; exposed via REST API.
- Designed 38MB GPT-2 (PyTorch) for edge inference (perplexity: 20); published as `sagar-neuron-gpt2`. [Link](#)

Radiology Connect — Java Spring Boot, MySQL, MongoDB, ReactJS, JWT [Link](#) Dec 2024

- Engineered a secure teleradiology platform enabling remote image access and storage via Spring Boot and React.
- Designed dual-database architecture with MySQL for structured records and MongoDB for imaging data.
- Implemented JWT auth and real-time messaging for encrypted doctor–radiologist communication.

TeleManas Project (Ongoing) — Streamlit, LangChain, LangGraph, LoRA, MySQL, SQL [Link](#) Jan 2025 – Present

- Ongoing government-funded project under college for the national TeleManas mental health initiative.
- Building an intelligent dashboard that converts natural language queries into visual insights using LangChain.
- Automates data analysis to address policy-related queries from Government of India stakeholders.

PUBLICATIONS

Predicting Soil pH using Mobile Captured Picture [Link](#) March 2021
Gangasagar HL, Jovin Dsouza, Bhagyashree B Yargal, Arun Kumar SV, Anuradha Badage. *International Research Journal of Engineering and Technology (IRJET)*, Vol. 08, Issue 07, July 2021.

TECHNICAL SKILLS

Languages: Python, C, C++ **Technologies/Frameworks:** TensorFlow, PyTorch, Docker, Git, GitHub, GitHub Actions, Flask, MySQL, SQL, ML Flow, Apache Airflow, Raspberry Pi

RELEVANT COURSEWORK

Software Systems Concrete Mathematics Machine Learning Mathematics For ML Enterprise System Development
Visual Recognition Multi Objective Machine Learning Project Elective

LEADERSHIP / EXTRACURRICULAR

Volunteered in institute-level activities and events under the Aikyam student body.

Spring 2020 – Present

District Level Table Tennis Winner.